

SPIM Lectures on Fundamental Groups

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Covering maps and Lifting

Definition 1 : A map $p : \tilde{X} \rightarrow X$ is called a *covering map* if for all $x \in X$ there is an open neighbourhood U_x of x in X such that $p^{-1}(U_x) = \bigsqcup_{\alpha} U_{\alpha}$ where U_{α} 's are open in $\tilde{X}$ and the restriction map $p|_{U_{\alpha}} : U_{\alpha} \rightarrow U_x$ is a homeomorphism for all α 's.

Remark/Definitions : Such a neighbourhood U_x in the above definition is called an *evenly covered neighbourhood*. So the above definition says, essentially, that every point of X has an evenly covered neighbourhood. The homeomorphic copies U_{α} 's are called *slices above* U_x .

Theorem 2 : Let X, Y be topological spaces, $p : \tilde{X} \rightarrow X$ a covering map and $F : Y \times [0, 1] \rightarrow X$ be a map. Then given a lift $\tilde{F} : Y \times \{0\} \rightarrow \tilde{X}$ of $F|_{Y \times \{0\}}$, there is a unique lift $\tilde{F} : Y \times [0, 1] \rightarrow \tilde{X}$ of F that restricts to the given $\tilde{F}$ on $Y \times \{0\}$.

Proof. Fix a point $y_0 \in Y$. For every $t \in [0, 1]$, the point $F(y_0, t)$ will have an evenly covered neighbourhood U . Take a product neighbourhood $N_t \times (a_t, b_t)$ of (y_0, t) whose image under F lies in U . By compactness of $[0, 1]$ there are finitely many product neighbourhoods $N_1 \times [0, b_1], N_2 \times (a_2, b_2), \dots, N_k \times (a_k, 1]$ such that the intervals $[0, b_1], \dots, (a_k, 1]$ cover $[0, 1]$.

Now let $N = \bigcap_{j=1}^k N_j$ and throw away the redundant intervals from the cover of $[0, 1]$ to get a partition $0 = t_0 < t_1 < \dots < t_m = 1$ such that $F(N \times [t_i, t_{i+1}])$ is contained in some evenly covered neighbourhood, say U_i . Suppose we have already constructed $\tilde{F}$ on $N \times [0, t_i]$, given $\tilde{F}$ on $N \times \{0\}$ (the base case for induction). Since $F(N \times [t_i, t_{i+1}]) \subset U_i$ there is some slice $\tilde{U}_i$ above U_i containing $\tilde{F}_i(y_0, t_i)$, as $p\tilde{F}(y_0, t_i) = F(y_0, t_i) \in U_i$. But $\tilde{F}(N \times \{t_i\})$ may not lie completely inside $\tilde{U}_i$. So we may replace N by a smaller neighbourhood. Precisely, we may replace N by $N \cap (\tilde{F} \circ \iota)^{-1}(\tilde{U}_i)$, where $\iota : N \rightarrow N \times \{t_i\}$ is the natural inclusion. We continue to denote this smaller neighbourhood by N . Now we have $\tilde{F}(N \times \{t_i\}) \subset \tilde{U}_i$. Then on $N \times [t_i, t_{i+1}]$ define $\tilde{F} := p|_{\tilde{U}_i}^{-1} \circ F|_{N \times [t_i, t_{i+1}]}$. This completes the induction step. We have now constructed a lift $\tilde{F}$ on $N \times [0, 1]$. What happens when two such neighbourhoods N_1 and N_2 overlap? The lifts constructed on $N_1 \times [0, 1]$ and $N_2 \times [0, 1]$ must agree on the overlapping strip. This forces us to consider the case when Y is a point space. We omit Y from the notations. So suppose $\tilde{F}_1 : [0, 1] \rightarrow \tilde{X}$ and $\tilde{F}_2 : [0, 1] \rightarrow \tilde{X}$ are two lifts of $F : [0, 1] \rightarrow X$ such that $\tilde{F}_1(0) = \tilde{F}_2(0)$. As before pick a partition $0 = t_0 < t_1 < \dots < t_m = 1$ such that $F([t_i, t_{i+1}])$ lies inside some evenly covered neighbourhood, say U_i . Assume inductively that $\tilde{F}_1 = \tilde{F}_2$ on $[0, t_i]$. Now $\tilde{F}_1(t_i) = \tilde{F}_2(t_i)$ lie in some slice $\tilde{U}_i$ above U_i . But as $[t_i, t_{i+1}]$ is connected so are $\tilde{F}_1([t_i, t_{i+1}])$ and $\tilde{F}_2([t_i, t_{i+1}])$ hence both these images must lie completely inside $\tilde{U}_i$. By injectivity of p we have $\tilde{F}_1 = \tilde{F}_2$ on $[t_1, t_{i+1}]$ thus completing the induction step. This resolves the overlapping problem. Patching the lifts on each such $N \times [0, 1]$ we get a lift on $Y \times [0, 1]$. The uniqueness follows from its uniqueness on each $\{y\} \times [0, 1]$. $\square$

We have the following immediate corollaries.

Path Lifting : Given a path $f : [0, 1] \rightarrow X$ with $f(0) = x_0$ and a fixed lift $\tilde{x}_0 \in p^{-1}(x_0)$ of the point x_0 , there is a unique lift $\tilde{f} : [0, 1] \rightarrow \tilde{X}$ of f satisfying $\tilde{f}(0) = \tilde{x}_0$.

Proof. Take Y as a point space. The proof is already done above. $\square$

Homotopy Lifting : Given a path homotopy $f_t : [0, 1] \rightarrow X$ with $f_t(0) = x_0$ and a fixed lift $\tilde{x}_0 \in p^{-1}(x_0)$ of x_0 , there is a unique lift of f_t to a path homotopy $\tilde{f}_t : [0, 1] \rightarrow \tilde{X}$ satisfying $\tilde{f}_t(0) = \tilde{x}_0$.

Proof. Take $Y = [0, 1]$. The path homotopy f_t defines a continuous map $F : [0, 1] \times [0, 1]$ given by $F(s, t) := f_t(s)$. We get a unique lift $\tilde{f}_0 : [0, 1] \rightarrow \tilde{X}$ of f_0 such that $\tilde{f}_0(0) = \tilde{x}_0$. This gives a lift $\tilde{F} : [0, 1] \times \{0\} \rightarrow \tilde{X}$ defined as $\tilde{F}(s, 0) := \tilde{f}_0(s)$. We have a unique lift $\tilde{F} : [0, 1] \times [0, 1] \rightarrow \tilde{X}$ of F not only w.r.t the given partial lift on $[0, 1] \times \{0\}$ but also w.r.t the fixed lift $\tilde{x}_0$ of the point x_0 . We still have to show $\tilde{F}$ gives a path homotopy. The restrictions $\tilde{F}|_{\{0\} \times [0, 1]}$ and $\tilde{F}|_{\{1\} \times [0, 1]}$ give paths in $\tilde{X}$ which are lifts of $f_t(0)$ and $f_t(1)$ and start at $\tilde{F}(0, 0)$ and $\tilde{F}(1, 0)$ and hence these must be constant paths by uniqueness of path lifting. Hence $\tilde{F}$ gives a path homotopy. $\square$

Now we define $p = s \mapsto (\cos 2\pi s, \sin 2\pi s) : \mathbb{R} \rightarrow S^1$, $\tilde{\omega}_n = s \mapsto ns : [0, 1] \rightarrow \mathbb{R}$ and consider the loops $\omega_n := p \circ \tilde{\omega}_n$ based at $(1, 0)$.

Theorem 3 : $\pi_1(S^1, (1, 0)) \cong \mathbb{Z}$.

Proof. We show that $\Phi = n \mapsto [\omega_n] : \mathbb{Z} \rightarrow \pi_1(S^1, (1, 0))$ is an isomorphism of groups. First of all it's a homomorphism since $\Phi(m + n) = [\omega_{m+n}] = [p \circ \tilde{\omega}_{m+n}] = [p \circ (\tilde{\omega}_m * (\tau_m \circ \tilde{\omega}_n))] = [\omega_m * \omega_n] = [\omega_m] * [\omega_n] = \Phi(m) * \Phi(n)$ (here τ_m is translation by m).

Next to show that Φ is surjective, take $[f] \in \pi_1(S^1, (1, 0))$. Then f has a unique lift to a path in $\mathbb{R}$, say $\tilde{f} : [0, 1] \rightarrow \mathbb{R}$ s.t $\tilde{f}(0) = 0$ (i.e a path starting at 0). Then $n := \tilde{f}(1) \in \mathbb{Z}$ since $p\tilde{f}(1) = f(1) = (1, 0)$ hence $\tilde{f}(1) \in p^{-1}((1, 0)) = \mathbb{Z}$. But as $\mathbb{R}$ is convex $\tilde{f}$ is path homotopic to $\tilde{\omega}_n$. Then $[f] = [p \circ \tilde{f}] = [p \circ \tilde{\omega}_n] = [\omega_n] \in \text{im}(\Phi)$.

Finally to show that Φ is injective let $[\omega_n] = [\omega_m]$ i.e there is a path homotopy $f_t : [0, 1] \rightarrow S^1$ such that $f_0 = \omega_n$ and $f_1 = \omega_m$. Then there is a unique lift of this path homotopy to a path homotopy $\tilde{f}_t : [0, 1] \rightarrow \mathbb{R}$ such that $\tilde{f}_t(0) = 0$. But by uniqueness of path lifting $\tilde{f}_0 = \tilde{\omega}_n$ and $\tilde{f}_1 = \tilde{\omega}_m$ and hence $n = m$. This completes the proof. $\square$

Some applications of the fundamental group of S^1

Brouwer's Fixed Point Theorem : Every continuous map $h : \mathbb{D}^2 \rightarrow \mathbb{D}^2$ has a fixed point.

Proof. Suppose on the contrary for all $x \in \mathbb{D}^2$ we have $h(x) \neq x$. Define $r : \mathbb{D}^2 \rightarrow S^1$ as follows : define $r(x)$ to be the point on the boundary of $\mathbb{D}^2$ given by intersecting the ray starting at $h(x)$ and containing x . Then $r|_{S^1} = \text{id}_{S^1}$. Take a non-nullhomotopic loop $f_0 : [0, 1] \rightarrow S^1$ based at $(1, 0)$ (for example $f_0 = \omega_1$). Then this loop f is nullhomotopic in $\mathbb{D}^2$. Let $f_t : [0, 1] \rightarrow \mathbb{D}^2$ be such a null homotopy. Then $r \circ f_t : [0, 1] \rightarrow S^1$ is a nullhomotopy in S^1 with $r \circ f_0 = f_0$. This contradicts that $\pi_1(S^1)$ is nontrivial. $\square$

Borsuk-Ulam Theorem : For every continuous map $f : S^2 \rightarrow \mathbb{R}^2$ there is a pair of antipodal points x and $-x$ in S^2 such that $f(x) = f(-x)$.

Proof. Suppose not. Then consider the map $g = x \mapsto \frac{f(x) - f(-x)}{\|f(x) - f(-x)\|} : S^2 \rightarrow S^1$. Define a loop $\eta : I \rightarrow S^2$ by $\eta = s \mapsto (\cos 2\pi s, \sin 2\pi s, 0)$ winding around the equator once and consider the loop $h := g \circ \eta : I \rightarrow S^1$. Now $g(-x) = -g(x)$ so for $s \in [0, \frac{1}{2}]$, $h(s + \frac{1}{2}) = g(\eta(s + \frac{1}{2})) = g(-\eta(s)) = -h(s)$. Lift h to a path $\tilde{h} : I \rightarrow \mathbb{R}$ and note that $p\tilde{h}(s + \frac{1}{2}) = h(s + \frac{1}{2}) = -h(s) = -p\tilde{h}(s)$ i.e $(\cos 2\pi\tilde{h}(s + \frac{1}{2}), \sin 2\pi\tilde{h}(s + \frac{1}{2})) = (-\cos 2\pi\tilde{h}(s), -\sin 2\pi\tilde{h}(s))$. This implies the equations

$$\cos \pi(\tilde{h}(s + \frac{1}{2}) + \tilde{h}(s)) \cos \pi(\tilde{h}(s + \frac{1}{2}) - \tilde{h}(s)) = 0 = \sin \pi(\tilde{h}(s + \frac{1}{2}) + \tilde{h}(s)) \cos \pi(\tilde{h}(s + \frac{1}{2}) - \tilde{h}(s))$$

In any case one has $\tilde{h}(s + \frac{1}{2}) - \tilde{h}(s) = \frac{q(s)}{2}$, where $q(s)$ is an odd integer depending on s . But being continuous, q is a constant function. Abusing notation we denote this constant value of the function by q itself. Hence we have $\tilde{h}(1) = \tilde{h}(\frac{1}{2}) + \frac{q}{2} = \tilde{h}(0) + q$. Hence h is a loop that winds q many times around S^1 and so represents a nontrivial class in $\pi_1(S^1)$. On the other hand as $\pi_1(S^2)$ is trivial η is null homotopic in S^2 and hence $h = g\eta$ is null homotopic in S^1 , a contradiction. This completes the proof. $\square$

Lusternik-Schnirelmann Theorem : Let A_1, A_2, A_3 be closed subsets of S^2 and suppose they cover S^2 . Then at least one of these three sets will contain a pair of antipodal points.

Proof. For each A_i define $d_i = x \mapsto \inf_{y \in A_i} \|x - y\| : S^2 \rightarrow \mathbb{R}$. Define $f : S^2 \rightarrow \mathbb{R}^2$ as follows : $f(x) = (d_1(x), d_2(x))$. By Borsuk-Ulam theorem there is some $x \in S^2$ such that $d_1(x) = d_1(-x)$ and $d_2(x) = d_2(-x)$. If one of these are zero, say $d_1(x) = d_1(-x) = 0$ then both $x, -x \in A_1$. Otherwise both $x, -x \in A_3$. This completes the proof. $\square$

Ham-Sandwich Theorem : Let R_1, R_2, R_3 be three connected open regions in $\mathbb{R}^3$, each of which is bounded with finite volume. Then there is a plane which cuts each of them in half by volume.

Proof. At first consider a large ball about origin that encloses the three regions. Fix $x \in S^2$ and consider the family of planes perpendicular to the radial line joining the origin and x . Then there is a unique such plane that cuts the region R_i in two halves of same volume. Let $d_i(x)$ be the directed distance of this unique plane from the origin, calling it positive if the plane lies in the same side of the origin as x and negative if the plane lies in the opposite side. Then the functions d_i are continuous and satisfy $d_i(-x) = -d_i(x)$. Let $f : S^2 \rightarrow \mathbb{R}^2$ be defined as $f(x) = (d_1(x) - d_2(x), d_1(x) - d_3(x))$. Then by Borsuk-Ulam theorem there is some $y \in S^2$ such that $f(y) = f(-y)$ i.e $(d_1(y) - d_2(y), d_1(y) - d_3(y)) = (d_2(y) - d_1(y), d_3(y) - d_1(y))$. Hence $d_1(y) = d_2(y) = d_3(y)$. This completes the proof. $\square$

$\pi_1(S^n) = 1$ for $n > 1$ and induced homomorphisms

Recall that for a map $f : (X, x_0) \rightarrow (Y, y_0)$ between pointed topological spaces, we have the induced homomorphism $\pi_1(f) : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ defined as $\pi_1(f)([\alpha]) = [f \circ \alpha]$. It is easy to check that if $g : (Y, y_0) \rightarrow (Z, z_0)$ then $\pi_1(g \circ f) = \pi_1(g) \circ \pi_1(f)$ and $\pi_1(\text{id}) = \text{id}$. Hence if f is a homeomorphism, then $\pi_1(f)$ is an isomorphism of groups.

Lemma 1 : If a space X retracts onto a subspace A , then $\pi_1(\iota) : \pi_1(A, a) \rightarrow \pi_1(X, a)$, induced by the inclusion $\iota : (A, a) \rightarrow (X, a)$, is injective. If A is a deformation retract of X , then $\pi_1(\iota)$ is an isomorphism.

Proof. If $r : X \rightarrow A$ is a retraction then $r \circ \iota = \text{id}_A$ as $r|_A = \text{id}_A$. So $\pi_1(r\iota) = \pi_1(r)\pi_1(\iota) = \text{id}_{\pi_1(A, a)}$ and hence $\pi_1(\iota)$ is injective. If A is a deformation retract of X i.e there is a homotopy $r_t : X \rightarrow X$ s.t $r_0 = \text{id}_X$, $r_1(X) = A$ and $r_t|_A = \text{id}_A$ for all t . So if $f : I \rightarrow X$ is a loop based at $a \in A$, the composition $r_t \circ f$ gives a path homotopy from f to a loop $g := r_1 \circ f$ in A . So we have $\pi_1(\iota)([g]_A) = [\iota \circ g]_X = [g]_X = [f]_X$. Hence $\pi_1(\iota)$ is surjective. Finally if r_t is such a family then by abuse of notation we can take $r_1 : X \rightarrow A$ such that $r_1|_A = \text{id}_A$ i.e $r_1 \circ \iota = \text{id}_A$ i.e A is a retract of X as well. Hence $\pi_1(\iota)$ is injective and so an isomorphism. $\square$

Remark : The Lemma implies that S^1 can't be a retract of $\mathbb{D}^2$ as otherwise the inclusion $\iota : S^1 \rightarrow \mathbb{D}^2$ induces an injection $\pi_1(S^1) \rightarrow \pi_1(\mathbb{D}^2)$. This gives another proof of the Brouwer's fixed point theorem.

Lemma 2 : Let $X = \bigcup_{\alpha} A_{\alpha}$ where A_{α} 's are path connected open sets in X , $x_0 \in \bigcap_{\alpha} A_{\alpha}$, and each intersection $A_{\alpha} \cap A_{\beta}$ be path connected. Then every loop in X based at x_0 is path homotopic to a product of loops each of which are contained in a single A_{α} .

Proof. Take a loop $f : I \rightarrow X$ based at x_0 . Then each $s \in I$ has an open interval say $V_s \subset I$ such that $f(\overline{V_s}) \subset A_{\alpha}$ for some α . By compactness of I there are finitely many such V_s 's that cover I . We partition I by the endpoints of these finitely many V_s 's, $0 = s_0 < \dots < s_m = 1$. Then $f([s_{i-1}, s_i]) \subset A_i$ for some A_i . Then $f_i := f|_{[s_{i-1}, s_i]}$ is a path in A_i and $f = f_1 * \dots * f_m$. Now as $A_i \cap A_{i+1}$ is path connected we can pick a path g_i in $A_i \cap A_{i+1}$ from x_0 to $f(s_i)$. Now consider the loop $(f_1 * \bar{g}_1) * (g_1 * f_2 * \bar{g}_2) * (g_2 * f_3 * \bar{g}_3) * \dots * (g_{m-1} * f_m)$ which is path homotopic to f . Note that these loops lie in $A_1, A_2, \dots, A_m$ respectively. This completes the proof. $\square$

Theorem 3 : $\pi_1(S^n) = 1$ for $n > 1$.

Proof. Let $S^n = A_1 \cup A_2$ where A_1, A_2 are homeomorphic to $\mathbb{R}^n$ and $A_1 \cap A_2$ is homeomorphic to $S^{n-1} \times \mathbb{R}$. If $n > 1$, then $A_1 \cap A_2$ is path connected as $S^{n-1} \times \mathbb{R}$ is path connected. By the above theorem every loop in S^n based at $x_0 \in A_1 \cap A_2$ is path homotopic to a product of loops in A_1 or A_2 . Since A_1, A_2 are homeomorphic to $\mathbb{R}^n$, these loops are nullhomotopic. Hence any loop in S^n based at x_0 is nullhomotopic. $\square$

Corollary 4 : $\mathbb{R}^2$ is not homeomorphic to $\mathbb{R}^n$ for $n \neq 2$.

Proof. Let $n > 2$ (the argument for $n = 1$ is well known). Then for $x \in \mathbb{R}^n$, $\mathbb{R}^n \setminus \{x\} \cong \mathbb{R}^n \setminus \{0\} \cong S^{n-1} \times \mathbb{R}$. Hence $\pi_1(\mathbb{R}^n \setminus \{x\}) \cong \pi_1(S^{n-1}) \times \pi_1(\mathbb{R}) \cong 1$. But $\pi_1(\mathbb{R}^2 \setminus \{0\}) \cong \pi_1(S^1) \cong \mathbb{Z}$. $\square$

Remark : In the proof above we have used the fact that $\pi(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$. The isomorphism is given by $[\varphi] \mapsto ([p_X \circ \varphi], [p_Y \circ \varphi])$ with inverse $([\alpha], [\beta]) \mapsto [(\alpha, \beta)]$.